PROJECT REPORT:
‘F1 The Movie’ Sentiment Analysis

1) Problem Statement

Movie reviews are unstructured text data. This project aims to:

- Convert text data into numerical features
- Train machine learning models
- Predict sentiment categories
- Compare performance of different models

---

2) Objective

The objective of this project is to analyze audience sentiment for the movie **“F1 the Movie”** using machine learning techniques.
The system classifies movie reviews into:

- Positive
- Neutral
- Negative

This helps in understanding public opinion and overall reception of the movie.

---

3) Dataset

- Total Reviews: **100**
- Source: Manually created simulated movie reviews
- Format:
    - review → text content of movie review
    - sentiment → label (positive / neutral / negative)

Dataset Split:

- 80% Training Data
- 20% Testing Data

---

4) Methodology

1. Data Preprocessing

- Converted all text to lowercase
- Removed punctuation and special characters
- Cleaned noisy text data

2. Feature Extraction

- Used **CountVectorizer (Bag of Words model)**
- Converted text into numerical vectors

3. Machine Learning Models Used

Naive Bayes

- Probabilistic classifier
- Fast and efficient for text classification

Logistic Regression

- Linear classification model
- Performs well on structured feature spaces

4. Model Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score

A confusion matrix was also generated for visualization.

---

5) Results

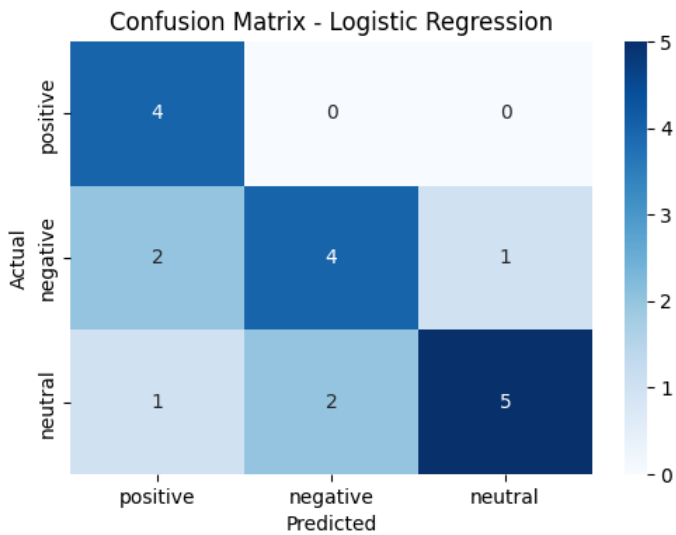
Model	Performance
Naive Bayes	Moderate accuracy
Logistic Regression	Better overall performance

---

6) Key Insights

- Logistic Regression performed better than Naive Bayes.
- Neutral reviews are hardest to classify correctly.
- Bag of Words gives basic but limited feature understanding.

- Model performance is limited due to small dataset (100 samples).
- Accuracy can improve with TF-IDF and more data.




```
PS C:\Users\kkaif\OneDrive\Desktop\30_KashafKhan_Assignment2> python notebook/sen
timent_analysis.py

==============================
NAIVE BAYES RESULTS
==============================
              precision    recall  f1-score   support

    negative       1.00      0.57      0.73         7
     neutral       0.83      0.62      0.71         8
    positive       0.44      1.00      0.62         4

    accuracy                           0.68        19
   macro avg       0.76      0.73      0.69        19
weighted avg       0.81      0.68      0.70        19

Accuracy: 0.6842105263157895

==============================
LOGISTIC REGRESSION RESULTS
==============================
              precision    recall  f1-score   support

    negative       0.67      0.57      0.62         7
     neutral       0.83      0.62      0.71         8
    positive       0.57      1.00      0.73         4

    accuracy                           0.68        19
   macro avg       0.69      0.73      0.69        19
weighted avg       0.72      0.68      0.68        19

Accuracy: 0.6842105263157895
```

7) How to Run

```
pip install -r requirements.txt
python sentiment_analysis.py
```

---

8) Conclusion

Logistic Regression is the better-performing model for this dataset.
However, performance is limited due to:

- Small dataset size (100 samples only)
- Basic feature extraction method

---

9) Future Improvements

- Increase dataset size
- Use TF-IDF instead of Bag of Words
- Apply advanced models like SVM or BERT
- Hyperparameter tuning for better accuracy

---

10) Tools And Libraries Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

---

11) Student's Details

Name: KASHAF KHAN
Roll No: 30
UIN: 231A055
YEAR: TE-AIDS